جامعة الإمام عبد الرحمن بن فيصل
IMAM ABDULRAHMAN BIN FAISAL UNIVERSITY
College of Business Administration

Final Project

Digital Transformation in Saudi Arabia's Financial Sector: A Comparative Study of Al Rajhi Bank and STC Pay .

Subject: Digital Business

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1. Introduction:

Saudi Arabia is on a rapid transition to being a digital economy with its rapid development of digital infrastructure and digital technology in support of Vision 2030. The rapid growth of digital technology and e-commerce has created new and exciting opportunities for new operating models and digital business strategies. This paper will discuss how digital technology and e-commerce will affect the transformation of the Saudi business sector.

The purpose of this research study is to investigate the ways digital technologies and e-commerce can be leveraged to promote the transformation of the Saudi Arabian business sector. The research will include identifying the key areas of digital technology development, business transformations that occur because of those developments, and identifying the barriers and challenges that must be addressed in order to leverage digital technologies. The research will provide a framework for businesses to implement digital technology, promote innovation and improve customer experience through the integration of digital technologies into their respective business models.

The research will utilize case studies of companies in the Saudi Arabian marketplace that have been successful in implementing digital technology and e-commerce solutions, such as Al Rajhi Bank's implementation of fintech services and STC Pay's disruption of traditional banking services. Additionally, the research will discuss how to adapt to the ever-changing Saudi Arabian business environment, the development of solutions for overcoming barriers and challenges, and the opportunity to leverage digital technology to continue to grow and thrive in a digital economy.

2. Literature Review

1-Business Models & Digital Commerce: Important Models/Theories:

Definitions: Digital business models utilize digital platforms to offer value, create revenue, and deliver products/services. Examples of models include e-commerce and digital platform-based business models.

Frameworks: Business Model Canvas: a framework to analyze what structure to use and how the digital business operates from a value proposition, target customers, and so on.

Disruptive Innovation (Clayton Christensen): An example of a Disruptor idea, the contrast between the way traditional banks work and how disruptors operate; One example of a new technology company disrupting this is FinTech Start Up, STC Pay has provided improved access to traditional banks.

Combined Example (including the above) Al Raji Banking: Traditional banking combined with digital innovation.

STC Pay has only platform-based business model FinTech (disrupting the traditional Financial Services)

Definition: FinTech is a technology-based business providing financial services including digital payment systems, lending and Investment Wealth Management, which is disruptive to traditional banking models.

What is the effect? Technology such as Mobile Payments, Blockchain, and Ai provides consumers access to managing Financial Transactions differently than

previously possible. Example, STC Pay provides digital wallets, mobile payments; this will provide greater disruption to the traditional banking model

Example:

- Al Rajhi Bank integrates digital banking services to enhance customer service.
- STC Pay offers digital payment solutions, driving the fintech revolution in Saudi Arabia.
- Models of E-Commerce and Digital Business: Theory & Framework.

2-Defining Digital Business Models:

Digital business models utilize platforms (online or offline) to deliver (distribute) value and generate (create) revenue for the business; this is normally done via eCommerce or other platform-based methods.
Key Frameworks of Digital Business Models:
Business Model Canvas: Describes the way companies operate in digital businesses. This includes examining the value of the proposition and identifying the customer segments.

Disruptive Innovation Theory (Clayton Christensen):

- Companies like STC Pay that offer services to consumers but are at the same time offering reduced costs to the consumer are examples of disruptions that affect traditional banks (e.g. STC Pay).
- Models of E-Commerce and Digital Business: Theory & Framework

Example:

- Al Rajhi Bank blends traditional banking with digital innovations.
- STC Pay operates entirely on a platform-based business model.

3-Fintech is Disruption of the Traditional Banking Industry:

Fintech uses technology to offer financial services (e.g., digital payment systems, loans, investment/wealth management) that disrupt the way traditional financial systems operate.
The impact includes Mobile Payment Systems, Blockchain Technology, and Artificial Intelligence All three of these trends are changing the way that consumers access and use Financial Services.

Example:

- STC Pay offers mobile payments and digital wallets, challenging traditional banking methods.
- Al Rajhi Bank adopts fintech solutions, such as mobile apps and blockchain, for better service delivery.
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3. Case Studies “STC Bank VS ALRAJHI Bank”:

Al Rajhi Bank's Digital Transformation Journey:

Al Rajhi Mobile App:

1- Al Rajhi Mobile is one of the best apps for digital banking services, with a usage rate exceeding 90%.

2- The app offers standard banking services along with additional features such as:

- Bill Payments
- Investment
- Apple Pay and Google Pay
- Reward points with the option to redeem them
- Financing Applications 3- Next-Generation ATMs:
- Previously, bank cards were created via ATMs, but now they can be created and activated through the Al Rajhi app.
- The ATMs offer interactive services such as phone assistance and interactive check withdrawals at branch ATMs.

4- Artificial Intelligence for Customer Support:

The automated assistant, accessible through Al Rajhi platforms and the bank's app, provides assistance with over 3 million customer inquiries annually.

Change at Al Rajhi Bank:

Location Strategy:

- Al Rajhi Bank focused on locating its branches in high-growth areas (usually commercial districts).
- Equipping all branches with smart devices to facilitate services, resulting in reduced paperwork and shorter customer service times.
- Establishing advanced consultation areas (moving away from the traditional approach) within the bank.

Results and their Impact on Al Rajhi Bank:

Digital Reliance:

- 95% of transactions are conducted through the Al Rajhi app.
- In 2018, Al Rajhi Bank had 2.5 million users, and by 2023, this number had grown to over 8 million.
- 70% of individuals use Al Rajhi Bank, reflecting the app's success in terms of ease of use and security.

Improved Customer Service:

- A 90% decrease in branch service rates is attributed to two main factors: technology and the increased number of Al Rajhi app users.

- 24/7 service for most banking services.
- Customer-specific offers are supported by calculations and analytics.

Success in several areas, including:

- Successfully reducing face-to-face services and transitioning them into digital transactions.
- Building a fully integrated digital interface.
- Cybersecurity: Establishing a multi-layered security system and maintaining a user-friendly interface.
- Compliance with financial regulations.

The Future:

- Integrated Finance: Financing partners such as companies to facilitate investment.
- Virtual Applications: Financial Literacy and education.
- Advanced Artificial Intelligence: Faster customer service and enhanced security.
- Expansion: Opening Al Rajhi Bank branches across various countries.

STC Pay: Analyze its model as a pure-play digital financial service:

Strategy:

STC Bank operates on the TechFin model, leveraging technology and infrastructure to deliver financial services. This strategy is based on: 1- Targeting the largest possible segment of the population

2- Intangible integration: Integrating financial services into everyday mobile phone usage, such as money transfers via mobile number

3- Integrated ecosystem: Previously, STC Pay was simply a mobile wallet, but today it has become a fully integrated digital bank.

Objectives of STC Pay's transformation into a fully integrated digital bank:

- Providing credit services
- Expanding account types, such as savings accounts
- Enhancing user trust and expanding financial services

Competitive Advantages:

- A ready customer base of over 26 million users.
- A seamless registration experience compared to traditional banks.
- Attractive pricing and free transfers, which have attracted a large number of users and merchants.
- Government-backed solutions: A key partner in the government's digital transformation.

How to benefit from STC's ecosystem:

- Its extensive customer base and data can be used to create suitable offers and reduce credit risks.
- Distribution Channels: Establishing multiple points of sale to cover different regions and segments of society.
- Billing and Marketing: Bill payments and marketing via the app and SMS.
- Merchant Relationships: Free transfers attract merchants to use STC as a payment platform.

Challenges and Risks:

1- Competition: Traditional banks are developing platforms similar to STC Pay, such as Al Rajhi Bank, and some global service providers like Apple Pay.
2- Regulation: With continued growth, it will be subject to increased regulatory oversight.
3- App and Features: Expanding services could potentially harm the bank; therefore, a balanced approach and careful consideration of each step are necessary.
4- Increased Difficulty: As STC Pay transforms into a bank, operations, risk management, and user experience will become more complex.

Vision:

STC Bank is not just a digital wallet (STC Pay), but a model designed to be an integral part of the user's life.

Comparison:

Comparison Aspect	STC Pay (STC Bank)	Al Rajhi Bank
Core Model / Overall Strategy	TechFin Model: Transition from telecom operator to financial service provider. Focus on "seamless integration" into the user's digital life.	FinTech from within a traditional bank: Deep digital transformation of an established financial institution. Focus on "technology serving banking."
Primary Target Customer	Mass market & the un/underbanked (youth, expatriates): Targeting financial inclusion through simplicity and mobile access.	Existing and banked customers: Deepening relationships with a massive existing customer base and attracting new digital-native clients

Starting Point & Foundation	Non-bank origin: STC's telecom customer base (>26 million). Trust derived from telecom brand.	Traditional banking origin: Extensive branch network and longstanding reputation as the world's largest Islamic bank. Trust derived from banking heritage.
Digital Strategy	Mobile-First, AppCentric: Built from scratch as a pure digital service; the app *is* the bank.	Omnichannel, Holistic Integration: Enhancing and integrating all channels (branches, nextgen ATMs, app) for a unified experience.
Role of Branches / Physical Presence	Limited & complementary: STC outlets used as "service points" to support digital operations (cash-in/cash-out, onboarding).	Central yet reimagined: Branches located in high-growth areas but made "smart" with advanced devices and consultation zones to support the digital journey.
Key Growth Drivers	1. Ecosystem Advantage: Integration with STC billing and data.	1. Converting existing customers: Moving 95% of transactions to the app.
	2. Attractive Pricing: Low/free transfers. 3. Government partnerships.	2. Enhanced experience: Ease of use and security. 3. Product innovation: Features like card issuance via the app.
Competitive Advantage	Speed, simplicity, and integration into daily life. Ability to penetrate segments traditionally overlooked by banks.	Credibility, depth in financial products (financing, investment), and strong financial infrastructure.
Results	- Transition to a full digital bank. - Leveraging a readymade user base.	- User growth: From 2.5M to 8M+ (via app). - Operational efficiency: 90% decrease in branch service rates. - Market dominance (used by 70% of individuals).

Main Challenges	1. Competition from both digital and traditional banks. 2. Increased complexity as it becomes a bank. 3. Stricter regulatory oversight.	1. Maintaining simplicity and agility amid massive scale. 2. Continuous innovation to keep pace with pure digital rivals. 3. Managing costs and transformation of legacy infrastructure (branches).
Future Vision	To become an inseparable part of the user's digital life, extending beyond banking to lifestyle services.	Integrated finance and global expansion: To be a comprehensive financial platform locally and globally, enhanced by financial literacy and AI.

Conclusion & Final Analysis:

- **Al Rajhi Bank** represents **"digital transformation from within"**: A financial giant reinvented its processes and customer experience using technology while retaining its core and financial strength. Its success lies in **converting existing customers** and maximizing efficiency.
- **STC Pay** represents **"disruption from the outside"**: A player from a completely different sector (telecom) entered with a simple, focused product and leveraged its non-financial assets to build a new financial institution from scratch. Its success lies in **attracting new customers** into the financial system and redefining the concept of banking.



Key Positioning:

- **Al Rajhi Bank** appears to straddle between "Legacy Players" and "Strategic Transformers," suggesting it maintains traditional banking strengths while undergoing significant digital transformation. The chart seems to show it positioned in the bottom-right area, indicating it has moved substantially toward digital-native approaches while maintaining elements of traditional banking.
- **STC Pay** is clearly positioned as a "Visionary Leader" in the top-right quadrant, reflecting its status as a purely digital, highly disruptive player that has redefined banking expectations in Saudi Arabia.

PERFORMANCE ANALYSIS:

1. **Mobile App Usage**: Both perform exceptionally well, with STC Pay achieving perfect digital penetration due to its app-only model.
2. **Digital Transactions**: STC Pay maintains 100% digital transactions (by design), while Al Rajhi Bank has successfully digitized 95% of transactions from a traditional starting point.
3. **Online Onboarding**: This metric shows the biggest gap (35 percentage points), highlighting STC Pay's superior digital-first customer acquisition model versus Al Rajhi's hybrid approach.

4. Opportunities

The impact of Payment Initiation Services (PIS) is also high on the players since the proposed opportunity, open banking, allows new business models outside of traditional card payments, including account-to-account payments through third-party apps, easier checkout, and new fintech integrations that enhance customer experience and efficiency of transactions.

The new e-commerce payment interface of Saudi Arabia (introduced 7 July 2025) opens a direct chance to enhance performance of online payments and the ability of merchants. It enables easier and secure online transactions of e-commerce and provides room to innovate merchant services, reconciliation, and payment acceptance by enhancing the payments ecosystem of online stores and gateways as well as the payment technical service providers.

Mada also does not only enhance the volume of transactions but also contributes to a more cashless economy as the global wallet is expanded. The introduction of Google Pay through made by SAMA, and the collaboration of Ant International to facilitate Alipay+ acceptance by 2026 could further speed up the adoption of contactless payments, reinforce the use of cross-border spending, and benefit merchants, particularly those operating in the tourism, retail, and high traffic services industries.

Lastly, financial inclusion programs can increase the target market of digital financial services. Revising account-opening regulations to add the concept of Visitor ID will enable visitors to open an account under specified circumstances, enabling a greater online payment uptake among non-resident groups and facilitating the expansion of the ecosystem in general, establishing new acquisition and onboarding channels not only in incumbent banks but also in digital banks.

5. Challenges:

The key dilemma of the digital transformation of financial sectors in Saudi Arabia is how to cope with rapid innovation, as well as strict regulatory adherence and risk management. With the expansion of digital frameworks by SAMA, including open banking (including Payment Initiation Services) and e-commerce payment infrastructure, the control mechanisms used by banks and FinTech's must ensure the security of consumers and the stability of the system despite the need to provide consumers with a quick product upgrade.

There is also the increasing pressure of cybersecurity and digital fraud since more transactions are moving to mobile space and API-based ecosystems are expanding. The movement to payment initiation, integrated merchant, and expanded connectivity of third parties adds to the attack surface and necessitates enhanced identity assurance, monitoring, and response to an incident by all involved parties. Data administration and assent administration in scale are still challenging. The open banking models require explicit customer consent, safe data exchange, and accountability among banks, FinTech's, and service providers; when consent is poorly executed or roles are ambiguous, trust levels diminish, and adoption tends to slow down.

Comparatively speaking, Al Rajhi Bank will have the transformation challenge of implementing changes in a large and complex company and maintaining platforms

that are unified and properly governed, particularly as it expands a fintech portfolio that also includes payments (neoleap), data aggregation (neotek) and personal finance management through Drahim. Meantime, the issue faced by STC Bank is the transitioning to a wallet/fintech background to full digital banks with bank-grade governance and controls without losing the ease and quickness of user experience that customers would prefer in a telecom-linked digital brand.

6. Conclusion:

This study examined digital transformation in Saudi Arabia financial sector through a comparative analysis of Al Rajhi Bank and STC Pay highlighting two distinct yet complementary transformation paths Al Rajhi Bank represents digital transformation from within a traditional financial institution, leveraging its strong banking heritage, regulatory expertise, and large customer base to migrate the majority of transactions to digital channels. Its success is reflected in high mobile app adoption operational efficiency, enhanced customer experience and the effective integration of advanced technologies such as artificial intelligence smart branches, and digital financing services.

In contrast, STC Pay illustrates disruption from outside the traditional banking system through a TechFin model. Originating from the telecommunications sector, STC Pay capitalized on its extensive customer ecosystem, data capabilities, and mobile-first design to rapidly penetrate the financial services market. By offering seamless onboarding, low-cost digital payments, and ecosystem-based services, STC Pay has contributed significantly to financial inclusion and redefined customer expectations of banking in Saudi Arabia.

The comparison demonstrates that Saudi Arabia's digital financial landscape benefits from both models: incumbent banks driving depth, trust, and product sophistication, and digital-native platforms driving speed, accessibility, and innovation However both face challenges related to regulatory compliance cybersecurity, data governance, and operational complexity. Overall, the findings confirm that digital transformation is not a single pathway but a strategic alignment of technology, regulation, and customer-centric design, supporting the Kingdom's Vision 2030 goals of financial inclusion innovation and a cashless economy.

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